

HARSHAL VANKUDRE

AI Engineer — Agentic Systems & Production RAG | Data Science Student
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PROFILE

AI engineer building reliable agentic systems and production RAG applications. Currently researching sandboxing and isolation for autonomous browser agents at Mercedes-Benz Tech Innovation. Previously sole AI developer at RÜKO Baumaschinen, taking a retrieval assistant from ingestion pipeline to daily use by ~50 staff.

PROFESSIONAL EXPERIENCE

Working Student – Agentic AI Security & Research Jun 2026 – Present
[Mercedes-Benz Tech Innovation GmbH | Stuttgart, Germany](#)

- Research secure execution environments — sandboxing, isolation, controlled resource access — for browser agents that run unattended
- Prototype sandbox and policy configurations, and assess NVIDIA OpenShell as one candidate agent runtime against them
- Document the security requirements a browser-agent workflow must meet to run robustly and reproducibly

Working Student – AI Development Oct 2025 – Apr 2026
[RÜKO GmbH Baumaschinen | Malsch, Germany](#)

- Sole developer for the company's AI solutions — requirements analysis through to production*
- Built and shipped a production RAG assistant on the OpenAI API, now in daily use by ~50 staff; tuned retrieval and prompting against real user questions
 - Built and operated the document-ingestion pipeline behind it, indexing structured and unstructured company data for vector search (Python, PostgreSQL, cloud-hosted)
 - Delivered a Microsoft Teams equipment assistant over 2,395 machines: FastAPI/LangGraph orchestration routing between SQL inventory lookups, Pinecone search across technical manuals, and Gemini advisory answers
 - Built an AI avatar-video pipeline (Higgsfield AI) generating listing videos for used-machine sales

Working Student – NLP & Automation Sep 2024 – Feb 2025
[EnBW Energie Baden-Württemberg AG](#)

- Co-developed an NLP customer-service chatbot and a GPT pilot that automated ~60% of tier-1 inquiries
- Automated recurring inquiry workflows; ran the supporting data, analysis, and reporting in HubSpot CRM and Excel

PROJECTS

WinMux — tiling terminal for agent fleets [TypeScript](#) · [Electron](#) · [ConPTY](#)
tmux-style tiling terminal for Windows that drives a fleet of parallel Claude Code sessions in one window, on real ConPTY shells.

Ember — AI coding usage & cost tracker [Rust](#) · [Tauri](#)
Desktop widget reading local Claude Code and Codex logs for live plan headroom, session burn, and true API-equivalent cost.

StudyOS — AI-generated study workspaces [Next.js](#) · [TypeScript](#) · [Prisma](#) · [Vercel AI SDK](#)
Turns one sentence about a student's courses into a populated semester workspace: planner, assignment board, reading list.

EDUCATION

B.Sc. Data Science Mar 2026 – Present
[Karlsruhe University of Applied Sciences \(HKA\)](#)

- Focus areas: machine learning, statistics, data engineering, and programming

Business Information Systems – Coursework Oct 2022 – Oct 2025
[Karlsruhe Institute of Technology \(KIT\)](#)

- Software development, databases, business processes, agile methods (Scrum); Java, Python, SQL, and data analysis — transferred to the practice-oriented Data Science program

Foundation Program – Business Track (W-Kurs) Oct 2021 – Jul 2022
[Konstanz University of Applied Sciences \(HTWG\)](#)

- Feststellungsprüfung passed — overall 2.5 | Computer Science 1.5 | English 1.3

TECHNICAL SKILLS

Agentic AI & Security	Browser Agents, Agent Sandboxing & Isolation, Policy Design, MCP Servers, Multi-Agent Workflows
AI & LLM Development	OpenAI API, Gemini, LangGraph, RAG Architecture, Vector Search, Ollama, LLM Fine-Tuning
Software Engineering	Python, FastAPI, TypeScript, JavaScript, Next.js, React, Node.js, Java, REST APIs
Data & Databases	PostgreSQL, MySQL, Pinecone, Redis, NoSQL, Pandas, Data Analysis
Tools & Platforms	Git, GitHub, Docker, Microsoft Azure, Jira, Linux

LANGUAGES

German	Fluent (C1)
English	Full professional proficiency (near-native)
Hindi	Native